

DETERMINATION OF HYDRAULIC COEFFICIENTS OF AN ORIFICE

AIM:-

To determine the hydraulics coefficients (C_d , C_v and C_c) of the given orifice and plot the graph Q_a Vs h and Q_a Vs $\sqrt{h}$.

APPARATUS USED

1. Orifice fitted in the orifice tank.
2. Collecting tank fitted with a piezometer
3. Stop watch.

PRINCIPLE

$$\text{Coefficient of discharge, } C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}}$$

$$\text{Theoretical discharge, } Q_{th} = a \sqrt{2 g h} \text{ cm}^3/\text{s}.$$

Where

$$a = \text{Area of orifice} = \frac{\pi d^2}{4} \text{ (d= diameter of orifice) in cm}^2$$

$$g = \text{Acceleration due to gravity.} = 981 \text{ cm/s}^2$$

$$h = \text{head above the centre of the orifice in cm.}$$

$$\text{Actual discharge, } Q_a = \frac{A \times R}{t} \text{ m}^3/\text{s}$$

Where

$$A = \text{Area of collecting tank in cm}^2$$

$$R = \text{Height of water collected in cm.}$$

t = Time for collection 'R' cm of water.

Coefficient of velocity,

$$C_v = \sqrt{\frac{X^2}{4Yh}} \quad \text{or} \quad \frac{X}{\sqrt{4Yh}}$$

X = Horizontal distance from the venacontracta to another point = $(X_2 - X_1)$ cm

Y = Vertical distance from the venacontracta to the point. = $(Y_1 - Y_2)$ cm

X_1 = Co-ordinate horizontal distance from the venacontracta

X_2 = Co-ordinate horizontal distance to the another point

Y_1 = Co-ordinate vertical distance from the venacontracta.

Y_2 = Co-ordinate vertical distance to the point.

$$\text{Coefficient of Contraction, } C_c = \frac{C_d}{C_v}$$

PROCEDURE

1. The diameter of the orifice and the internal plan area of the collecting tank are measured.
2. Start the pump and adjust the bypass valve to the orifice tank is regulated and water is allowed to fill the orifice tank to a constant head.
3. The outlet valve of the collecting tank is closed tightly and the time ' t ' required for 'R' meter rise of water is noted.
4. Set the hook gauge at the venacontracta and noted the reading X_1 & Y_1 .
5. The hook gauge is adjusted to the (flow is spread before) point take the reading X_2 & Y_2 .

The above procedure is repeated for different heads and the observations are tabulated the hydraulics coefficients.

[illegible][illegible]

RESULT

Coefficient of discharge

From

Calculation,

$C_d =$

Graph

$C_d =$

Coefficient of Velocity

$C_v =$

Coefficient of Contraction

$C_c =$

INFERENCE

GRAPHS

1. $Q_a \propto V_s \propto h$
2. $Q_a \propto V_s \propto \sqrt{h}$

Model graph of hydraulic coefficient of an orifice

